

Fallou Diouf

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[GitHub](#) | [Portfolio](#) | [LinkedIn](#)

PROFESSIONAL OBJECTIVE

Master's student in **Vision and Machine Intelligence** at Université Paris Cité, seeking a **4 to 6-month final-year internship starting in February 2027** in **Computer Vision and Deep Learning**. Interested in image processing, semantic segmentation, object detection, tracking, and visual recognition.

EDUCATION

2026 – 2027 | Université Paris Cité, Paris, France

Master 2 – Vision & Machine Intelligence

Key areas: Image Processing, Video Analysis, Pattern Recognition, 3D Imaging, Biomedical Imaging, Deep Learning, Machine Learning, Generative AI.

2025 – 2026 | Université Paris Cité, Paris, France

Master 1 – Vision & Machine Intelligence

2023 – 2024 | Université Gaston Berger, Saint-Louis, Senegal

Master 1 – Data Management and Software Engineering

Key areas: Artificial Intelligence, Advanced Algorithms, C/C++ Programming, Object-Oriented Programming, Databases.

2020 – 2023 | Université Gaston Berger, Saint-Louis, Senegal

Bachelor's Degree – Information Systems

ACADEMIC PROJECTS

2026 – 2027 | Remote Sensing Semantic Segmentation

Academic Project

Semantic segmentation of aerial images using a **Hypercolumns-based CNN**. Processed remote sensing images from the **Vaihingen dataset** and predicted a semantic class for each pixel.

Results: **70.88% Pixel Accuracy** and **38.71% Mean IoU**.

Skills: PyTorch, CNN, Semantic Segmentation, Computer Vision, Remote Sensing.

2025 – 2026 | Self-Supervised Test-Time Training

Pair Project

Implementation of a self-supervised learning approach combining **SimCLR** and a **Vision Transformer** to improve image classification robustness during inference.

Validation accuracy of **90.08%** and an improvement of **+0.39 percentage points** on CIFAR-10-C with Test-Time Training.

Skills: PyTorch, Vision Transformer, SimCLR, Self-Supervised Learning, Test-Time Training.

2026 – 2027 | Real-Time Multi-Person Tracking

Individual

Development of a multi-object tracking pipeline combining **YOLOv8** for person detection and **DeepSORT** for persistent identity tracking. Maintained person IDs across video frames and analyzed trajectories over time.

Results: 30 FPS input video, 360×640 resolution, up to 5 simultaneous tracks.

Skills: Computer Vision, Object Detection, Object Tracking, YOLOv8, DeepSORT, OpenCV, PyTorch.

2026 – 2027 | Face Mask Detection

Academic Project

Development of an object detection pipeline for identifying people with and without face masks using **Faster R-CNN** with a **ResNet-50 + FPN** backbone. Applied image preprocessing, data augmentation and model evaluation.

Skills: Object Detection, PyTorch, Faster R-CNN, ResNet-50, FPN, Transfer Learning.

2025 – 2026 | Coin Detection and Classification

Team & Individual

Development of a computer vision pipeline for detecting, counting and identifying euro coins from smartphone images. Applied Gaussian filtering, Canny edge detection and Hough Transform, combined with texture and diameter analysis.

Skills: Image Processing, Classification, OpenCV, Python, PyTorch, Feature Extraction.

TECHNICAL SKILLS

Computer Vision Semantic Segmentation, Object Detection, Object Tracking, Image Classification, Image Analysis, Shape Recognition, Video Analysis, OpenCV, YOLOv8, DeepSORT.

Deep Learning PyTorch, TensorFlow, CNN, ResNet, Faster R-CNN, U-Net, DeepLabv3, Swin-Unet, Vision Transformer, Transfer Learning, Self-Supervised Learning, SimCLR, Test-Time Training.

Machine Learning Scikit-learn, KNN, K-Means, PCA, Feature Extraction, Model Evaluation.

Programming & Tools Python, C++, C, SQL, NumPy, Pandas, Git, GitHub, Docker, Linux, FastAPI, Django.

PROFESSIONAL EXPERIENCE

Warehouse & Sorting Operative – ELIS

Jun 2026 – Aug 2026

Nanterre, France • Full-time

- Worked in sorting and order preparation operations within a fast-paced logistics environment.

Versatile Employee / Store Assistant – LP La Providence

Mar 2026 – Jun 2026

Paris, France • Part-time

- Maintained cleanliness and organization of a high-traffic retail environment.
- Worked independently while following hygiene, safety, and organizational procedures.

Programming Tutor – C/C++, Algorithms, OOP and SQL

Oct 2023 – Aug 2024

Université Gaston Berger, Senegal

- Supported students in C/C++, algorithms, object-oriented programming, and SQL through tutorials and practical sessions.
- Adapted technical explanations to individual difficulties and helped learners develop greater autonomy.

LANGUAGES: French: Full professional proficiency English: Intermediate

INTERESTS: Artificial Intelligence • Scientific Research • Computer Vision • Philosophical Discussions